

IM YU A HABAROVA, Russian Federation

WMO: 217210

Lat: 72.400N

Lon: 126.350E

Elev: 20

StdP: 101.09

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.3	(c) -39.4	(d) -43.7	(e) 0.1	(f) -41.3	(g) -41.8	(h) 0.1	(i) -39.2	(j) 22.5	(k) -28.2	(l) 20.1	(m) -29.7	(n) 2.4	(o) 180	(p) 1.114

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.1	(c) 21.9	(d) 15.9	(e) 18.8	(f) 14.0	(g) 15.9	(h) 12.4	(i) 16.4	(j) 21.1	(k) 14.5	(l) 18.3	(m) 12.7	(n) 15.7	(o) 5.3	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.0	19.1	12.3	8.9	17.1	10.7	8.0	14.9	46.0	21.4	40.8	18.0	36.0	15.7	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.3	(b) 16.3	(c) 14.3	(e) -42.5	(f) 25.4	(g) 2.5	(h) 3.8	(i) -44.3	(j) 28.2	(k) -45.8	(l) 30.4	(m) -47.2	(n) 32.6	(o) -49.0	(p) 35.3		
			DB													
			WB													
				(e) -42.5	(f) 17.9	(g) 2.4	(h) 1.7	(i) -44.2	(j) 19.2	(k) -45.6	(l) 20.2	(m) -46.9	(n) 21.2	(o) -48.7	(p) 22.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.7	-30.8	-29.4	-23.8	-15.7	-6.1	5.1	9.4	8.0	1.9	-9.3	-23.1	-27.7
	DBStd	15.63	4.73	6.05	7.66	6.91	4.96	5.73	4.64	3.61	3.78	5.70	5.85	5.77
	HDD10.0	8013	1265	1103	1047	771	499	169	68	86	246	598	994	1168
	HDD18.3	10966	1523	1337	1305	1021	757	397	284	321	494	857	1244	1426
	CDD10.0	95	0	0	0	0	0	23	47	23	2	0	0	0
	CDD18.3	6	0	0	0	0	0	1	5	0	0	0	0	0
	CDH23.3	51	0	0	0	0	0	10	41	0	0	0	0	0
	CDH26.7	11	0	0	0	0	0	1	9	0	0	0	0	0
Wind	WSAvg	6.4	7.6	7.8	6.6	5.9	5.5	5.5	5.5	5.3	5.4	6.6	7.1	8.1
Precipitation	PrecAvg	412	27	19	21	16	30	39	56	60	44	41	28	32
	PrecMax	608	71	66	45	59	64	71	128	127	106	67	53	85
	PrecMin	288	7	3	4	1	8	10	18	26	17	12	10	5
	PrecStd	74	15	14	11	13	14	17	26	23	21	16	12	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-16.3	-12.6	-3.0	1.7	9.0	24.5	27.9	21.0	14.8	1.3	-4.6	-13.5
		MCWB	-16.6	-13.0	-3.7	0.4	5.4	16.2	19.9	15.9	11.5	0.7	-5.1	-14.0
	2%	DB	-19.4	-16.0	-5.3	-1.2	5.2	20.6	23.5	18.0	10.4	0.3	-8.0	-15.8
		MCWB	-19.6	-16.3	-5.8	-2.2	3.2	14.8	16.9	14.3	8.7	-0.3	-8.5	-16.2
	5%	DB	-21.9	-19.5	-8.7	-3.2	2.9	17.2	20.1	15.8	8.6	-0.9	-12.0	-17.7
		MCWB	-22.1	-20.0	-9.2	-4.3	1.4	12.5	15.1	12.6	6.9	-1.6	-12.3	-18.0
	10%	DB	-24.0	-21.4	-11.3	-5.1	1.2	13.9	17.0	14.0	6.8	-1.9	-14.9	-19.3
		MCWB	-24.2	-21.7	-11.8	-6.0	0.2	10.7	13.4	11.6	5.4	-2.4	-15.3	-19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.7	-13.0	-3.5	0.2	5.7	17.0	20.1	16.3	11.4	1.0	-4.9	-13.9
		MCDB	-16.4	-12.6	-3.2	1.1	8.8	22.4	26.7	19.7	13.7	1.2	-4.5	-13.5
	2%	WB	-19.8	-16.3	-5.8	-2.2	3.5	14.9	17.6	14.6	8.9	-0.2	-8.4	-16.2
		MCDB	-19.5	-16.0	-5.3	-1.4	5.3	20.4	22.7	17.5	10.3	0.3	-7.9	-15.9
	5%	WB	-22.2	-19.8	-9.2	-4.2	1.5	12.9	15.4	13.0	7.2	-1.5	-12.3	-18.0
		MCDB	-22.0	-19.5	-8.7	-3.3	2.6	16.7	19.5	15.3	8.5	-1.0	-11.9	-17.7
	10%	WB	-24.2	-21.7	-11.8	-6.0	0.1	10.6	13.5	11.6	5.5	-2.4	-15.2	-19.6
		MCDB	-24.0	-21.5	-11.3	-5.1	1.0	14.0	17.1	14.0	6.6	-1.8	-14.9	-19.3
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.8	6.2	7.0	5.0	5.9	6.1	5.3	3.6	4.0	4.7	4.8
		MCDBR	7.0	7.0	8.0	8.0	6.6	12.0	10.3	8.8	5.9	3.8	7.8	7.1
		MCWBR	6.9	6.9	7.9	7.6	5.3	7.2	5.9	5.9	4.4	3.8	7.7	7.1
	5% WB	MCDBR	7.1	6.9	8.0	7.8	6.2	11.5	10.2	8.0	5.6	3.8	7.7	7.0
		MCWBR	6.9	6.8	7.9	7.5	5.2	7.4	6.5	5.8	4.6	3.8	7.6	7.0
Clear-Sky Solar Irradiance	taub	N/A	0.189	0.276	0.306	0.338	0.341	0.390	0.371	0.296	0.198	N/A	N/A	
	taud	N/A	1.945	2.042	2.088	2.202	2.442	2.320	2.337	2.418	1.977	N/A	N/A	
	Ebn at Noon	0	426	664	792	821	841	768	727	684	379	0	0	
	Edh at Noon	0	30	75	105	110	91	99	85	54	26	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.18	1.59	3.99	5.41	5.20	4.41	2.79	1.36	0.32	0.00	0.00	
	RadStd	0.00	0.02	0.09	0.17	0.29	0.45	0.35	0.32	0.14	0.04	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+1.05	N/A	N/A	N/A	N/A	N/A	-413	-394	N/A	N/A

Nomenclature: See separate page